KKK-LL TOP00, G01, and P01B2A.1.2

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KKK-LL TOP00 - Cauchy-Topology and Branch-Scope Lock

Canonical ID: TOP00
Roadmap: KKK-LL Harmonised Master Roadmap - P01B.1 No-Go Revision v3
Date: 10 August 2026
Status in this document: COMPLETE / FROZEN FOR DECLARED GOVERNANCE SCOPE
Direct prerequisites: C00 and P01B.1-FLAT
Downstream handoffs: P01B.2-CURV, P01B.3-GEN, P01B.M-FULL

1. Purpose

P01B.1-FLAT closes the following branch:

$$\text{flat} + \text{diagonal} + \text{homogeneous} + \text{metric-only } D = 16 + \text{persistent finite } 3 + 12 \text{ Hubble gap} + \text{full G6 rank}.$$

TOP00 prevents the revised programme from evading that result by silently changing the differentiable manifold, importing $(\mathbb{CP}^2)^3$ after the fact, or confusing stabilization on a prescribed product with dynamical creation of a product topology.

2. Global-hyperbolicity input

The classical branches governed by TOP00 assume a connected, smooth, time-oriented, globally hyperbolic Lorentzian spacetime (M^{16}, g). The Bernal-Sanchez splitting theorem then gives a smooth spacelike Cauchy hypersurface Σ^{15} and a global diffeomorphism

$$M^{16} \cong I \times \Sigma^{15}, \qquad g = -\beta(t,x)dt^2 + g_t,$$

for an interval $I \subseteq \mathbb{R}$. All Cauchy hypersurfaces in this class are diffeomorphic. Therefore smooth globally hyperbolic metric evolution changes g_t, not the diffeomorphism type of Σ^{15}.

Consequently, a transition

$$\mathbb{R}^{15} \longrightarrow \mathcal{X}^3 \times (\mathbb{CP}^2)^3$$

is **not** part of P01B.2-CURV. A topology-changing history would require a separately declared singular, non-globally-hyperbolic, cobordism, boundary, or quantum process and belongs to neither the present Cauchy problem nor the claims authorized by this lock.

3. Fixed topology for the curved product branch

Every P01B.2-CURV run must choose, before initial data are specified, one member of the following fixed-topology family:

$$\boxed{\Sigma^{15}_{\kappa,\Gamma} = \mathcal{X}^3_{\kappa,\Gamma} \times Y^{12}, \qquad Y^{12} = \mathbb{CP}^2_c \times \mathbb{CP}^2_L \times \mathbb{CP}^2_R.}$$

Here $\mathcal{X}^3_{\kappa,\Gamma} = \widetilde{\mathcal{X}}^3_\kappa/\Gamma$ is a chosen complete orientable constant-sectional-curvature three-manifold, with $\kappa \in \{-1, 0, +1\}$ after normalization. The topology and quotient group Γ are held fixed during a run. Different choices of (κ, Γ) are different branches; no solution is interpreted as evolving from one branch to another.

The three projective factors are already distinguished by the input labels c, L, R. When their radii and couplings coincide the equations must retain the permutation symmetry $\mathfrak{S}_3$, but the labels are not claimed to emerge dynamically.

4. Metric ansatz and dynamical content

P01B.2-CURV uses the multiply warped homogeneous product metric

$$\boxed{ds_{16}^2 = -N(t)^2 dt^2 + a(t)^2 d\Sigma_{3,\kappa}^2 + \sum_{A\in\{c,L,R\}} b_A(t)^2 ds^2_{\mathbb{CP}^2_A,\mathrm{FS},\, b=1}.}$$

The lapse N, external scale factor a, and three internal radii $b_c, b_L, b_R > 0$ are dynamical. The Hubble/radion rates are

$$H = N^{-1}\frac{\dot a}{a}, \qquad h_A = N^{-1}\frac{\dot b_A}{b_A}.$$

The ansatz fixes an orthogonal $3+4+4+4$ product distribution at the outset. It therefore tests:

1. existence of curvature-supported expanding/static-radius solutions;
2. stabilization and the relative selection of the three metric radii;
3. constraint propagation and full kinetic/principal-symbol health;
4. the basin of attraction, only after a P01B.2 pass.

It does **not** test:

1. creation of $(\mathbb{CP}^2)^3$ topology;
2. discovery of a $3+12$ split from fifteen unlabeled flat directions;
3. derivation of the labels c, L, R;
4. emergence of the Fubini-Study conformal class from a generic metric.

Those stronger questions belong to P01B.3-GEN and its later P02-P04 gates.

5. Action and sector firewall

The P01B.2-CURV branch is metric-only maximal Lovelock gravity:

$$S_{\mathrm{CURV}}[g] = \frac{1}{2\kappa_{16}^2}\int_{M^{16}} d^{16}x\sqrt{-g}\sum_{k=0}^{8}\alpha_k\mathcal{L}_k.$$

All local orders $\alpha_0,\dots,\alpha_7$ must be retained. The order $k=8$ density is the sixteen-dimensional Euler density. Its bulk Lovelock tensor vanishes identically, but its reduced action must be exhibited as a boundary/transgression term and retained in the action-level ledger.

No Yang-Mills field, fermion, G_6 flux, abelian internal flux, or externally prescribed stress tensor is permitted in P01B.2-CURV. Turning on any such sector defines P01B.M-FULL and changes the scientific claim.

6. Topological data fixed at entry

For each projective factor,

$$\dim_{\mathbb{R}}\mathbb{CP}^2 = 4, \quad \pi_1(\mathbb{CP}^2) = 0, \quad H^*(\mathbb{CP}^2;\mathbb{Z}) = \mathbb{Z}[H]/(H^3), \quad \chi(\mathbb{CP}^2) = 3.$$

Thus

$$H^*(Y;\mathbb{Z}) = \frac{\mathbb{Z}[H_c,H_L,H_R]}{(H_c^3,H_L^3,H_R^3)}, \qquad \chi(Y) = 27, \qquad \pi_1(Y) = 0.$$

The product is naturally Spin^c, while an ordinary spin structure is not silently assumed. These are immutable topological inputs. Continuous variables such as b_A, volumes, curvature scales, and scale hierarchies are metric data and remain dynamical.

7. Claim boundary by branch

Branch	Fixed input	Dynamical question	Claim explicitly forbidden
P01B.2-CURV	product topology and homogeneous product conformal classes	four scale factors, stabilization, full-rank health	topology or unlabelled split emergence

Branch	Fixed input	Dynamical question	Claim explicitly forbidden
P01B.3-GEN	compatible topology only	whether a generic metric develops a covariant $3+12$ distribution	importing a product metric or labels
P01B.M-FULL	topology plus complete matter/flux sector conventions	coupled compactification of the full action	calling success pure Lovelock
P01B.D-DAE	flat characteristic endpoint	reduced constraints/gauge/strong-coupling classification	automatic reopening of P01C-FLAT

8. P01B.2-CURV pass contract

P01B.2-CURV may authorize P01C-CURV only if it supplies all of the following:

1. the exact lapse constraint and four independent scale-factor equations with arbitrary $\alpha_0, \ldots, \alpha_7$;
2. the explicit α_8 boundary identity;
3. at least one finite-radius fixed point with $H_\star > 0$ and $h_{A\star} = 0$;
4. the complete homogeneous tensor/radion acceleration matrix inherited from G6, not merely the four-dimensional minisuperspace block;
5. a nonzero branch determinant and quantitative distance from every characteristic surface;
6. positive four-dimensional graviton, physical radion, and relevant vector residues after constraint/gauge reduction;
7. a stability analysis with the same declared numerical tolerance throughout;
8. exact symbolic regression against Einstein, EGB, equal-radius, single-factor, flat-curvature, and topological limits.

Failure of any one item blocks P01C-CURV. Fixed-point existence alone is not a pass.

9. Governing disposition

$$\boxed{\text{P01A}-\text{G6.1.1 FROZEN} \longrightarrow \text{P01B.1}-\text{FLAT COMPLETE/SCOPED NO}-\text{GO} \not\longrightarrow \text{P01C}-\text{FLAT.}}$$

$$\boxed{\text{TOP00} \longrightarrow \text{G01} \longrightarrow \text{P01B.2}-\text{CURV}}$$

TOP00 changes no Lovelock equation and rescues no flat solution. It only fixes the admissible manifold class and prevents overstatement of what the curved product calculation can establish.

10. References

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KKK-LL G01 - Exact Product Geometry of $(\mathbb{CP}^2)^3$

Canonical ID: G01
Role: direct prerequisite for P01B.2-CURV
Input convention: frozen G00 single-factor Fubini-Study normalization
Status in this document: COMPLETE / FROZEN FOR P01B.2 PREREQUISITE SCOPE

1. Single-factor data

For each factor $X_A = \mathbb{CP}^2_A$, $A \in \{c, L, R\}$, with radius $b_A > 0$, the G00 convention is

$$R_{a\bar{b}c\bar{d}} = \frac{2}{b_A^2}\left(g_{a\bar{b}}g_{c\bar{d}} + g_{a\bar{d}}g_{c\bar{b}}\right), \qquad \mathrm{Ric} = \frac{6}{b_A^2}g_A.$$

Consequently,

$$R_A = \frac{24}{b_A^2}, \qquad \mathrm{Vol}(X_A) = \frac{\pi^2 b_A^4}{2},$$

and, pointwise,

$$|\,\mathrm{Riem}_A\,|^2 = \frac{192}{b_A^4}, \qquad |W_A|^2 = \frac{96}{b_A^4}, \qquad \mathcal{L}_2[X_A] = \frac{192}{b_A^4}.$$

The integrated Lovelock densities are

$$\boxed{I_0^{(A)} = \frac{\pi^2 b_A^4}{2}, \qquad I_1^{(A)} = 12\pi^2 b_A^2, \qquad I_2^{(A)} = 96\pi^2, \qquad I_{p\geq 3}^{(A)} = 0.}$$

The last equality is dimensional: a four-manifold has no nonzero Lovelock density above order two. The value of I_2 also follows from $I_2 = (4\pi)^2 2!\,\chi(\mathbb{CP}^2)$.

2. Product curvature

For

$$Y^{12} = X_c \times X_L \times X_R, \qquad g_Y = g_c \oplus g_L \oplus g_R,$$

the Levi-Civita connection and Riemann tensor split block-diagonally:

$$\nabla^Y = \nabla^c \oplus \nabla^L \oplus \nabla^R, \qquad \mathrm{Riem}_Y = \mathrm{Riem}_c \oplus \mathrm{Riem}_L \oplus \mathrm{Riem}_R,$$

with all mixed curvature components zero. Hence

$$\mathrm{Ric}_Y = \bigoplus_A \frac{6}{b_A^2}g_A, \qquad R_Y = 24\sum_A \frac{1}{b_A^2}, \qquad \mathrm{Vol}(Y) = \frac{\pi^6}{8}b_c^4 b_L^4 b_R^4.$$

The product is Einstein if and only if

$$b_c = b_L = b_R.$$

Thus unequal radii are genuine shape moduli rather than a coordinate artifact.

3. Exact product Lovelock formula

The canonical Lovelock double-form product rule gives

$$\boxed{C_s(b_c, b_L, b_R) = \int_Y \mathcal{L}_s[g_Y]\,\mathrm{vol}_Y = \sum_{p+q+r=s} \frac{s!}{p!q!r!} I_p^{(c)} I_q^{(L)} I_r^{(R)}.}$$

Only $p, q, r \in \{0, 1, 2\}$ contribute, so $C_s = 0$ for $s > 6$. Writing $x_A = b_A^2$, the complete unequal-radius panel is

$$\frac{C_0}{\pi^6} = \frac{1}{8}x_c^2x_L^2x_R^2,$$

$$\frac{C_1}{\pi^6} = 3\left(x_cx_L^2x_R^2 + x_c^2x_Lx_R^2 + x_c^2x_L^2x_R\right),$$

$$\begin{aligned}\frac{C_2}{\pi^6} &= 24\left(x_c^2x_L^2 + x_c^2x_R^2 + x_L^2x_R^2\right)\\ &\quad + 144\left(x_c^2x_Lx_R + x_cx_L^2x_R + x_cx_Lx_R^2\right),\end{aligned}$$

$$\begin{aligned}\frac{C_3}{\pi^6} &= 1728\left(x_c^2x_L + x_c^2x_R + x_L^2x_c + x_L^2x_R + x_R^2x_c + x_R^2x_L\right)\\ &\quad + 10368x_cx_Lx_R,\end{aligned}$$

$$\frac{C_4}{\pi^6} = 27648(x_c^2 + x_L^2 + x_R^2) + 165888(x_cx_L + x_cx_R + x_Lx_R),$$

$$\frac{C_5}{\pi^6} = 3317760(x_c + x_L + x_R), \qquad \frac{C_6}{\pi^6} = 79626240.$$

Each expression is exactly $\mathfrak{S}_3$-invariant.

4. Equal-radius regression

For $b_c = b_L = b_R = b$, the panel reduces to the frozen C00 values:

s	$C_s(b)$
0	$\pi^6 b^{12}/8$
1	$9\pi^6 b^{10}$
2	$504\pi^6 b^8$
3	$20736\pi^6 b^6$
4	$580608\pi^6 b^4$
5	$9953280\pi^6 b^2$
6	$79626240\pi^6$

At top order,

$$C_6 = (4\pi)^6 6!\,\chi(Y) = (4\pi)^6 6!\cdot 27 = 79626240\pi^6,$$

which is an independent Chern-Gauss-Bonnet check.

5. Pointwise generating data for P01B.2-CURV

Because every factor is homogeneous, define the pointwise Lovelock data

$$(\ell_0^{(A)}, \ell_1^{(A)}, \ell_2^{(A)}) = \left(1, \frac{24}{b_A^2}, \frac{192}{b_A^4}\right)$$

and distinguish them from the exponential generating polynomial

$$P_A(z) = \sum_{p=0}^{2} \frac{\ell_p^{(A)}}{p!}z^p = 1 + \frac{24}{b_A^2}z + \frac{96}{b_A^4}z^2.$$

The pointwise product Lovelock densities then obey

$$\sum_{s=0}^{6} \frac{\mathcal{L}_s[Y]}{s!}z^s = \prod_{A=c,L,R} P_A(z),$$

where the coefficient convention includes the ordinary multinomial factor: equivalently,

$$\mathcal{L}_s[Y] = \sum_{p+q+r=s} \frac{s!}{p!q!r!} \ell_p^{(c)} \ell_q^{(L)} \ell_r^{(R)}, \quad (\ell_0, \ell_1, \ell_2) = \left(1, \frac{24}{b^2}, \frac{192}{b^4}\right).$$

Thus the factor 96 belongs to P_A because it is $\ell_2^{(A)}/2!$; it is not a change of the Lovelock normalization.

6. Scope boundary

G01 certifies the product tensors, topology, radii, and Lovelock invariants of the prescribed geometry. It does not establish that pure Lovelock dynamics selects this topology, this product conformal class, equal radii, a stable vacuum, a four-dimensional Planck sign, a gauge kinetic sign, chirality, or a consistent finite Kaluza-Klein truncation. Those are downstream gates.

7. Symbolic verification contract

The companion exact Python engine verifies, without floating-point arithmetic:

1. the real orthonormal Fubini-Study curvature tensor has $\mathrm{Ric} = 6g/b^2$, $R = 24/b^2$, and $\mathcal{L}_2 = 192/b^4$;
2. the double-form wedge calculation on the twelve-dimensional product agrees with the multinomial formula at every order $s = 0, \ldots, 6$;
3. the unequal-radius panel is permutation invariant;
4. the equal-radius panel reproduces C00;
5. the order-six coefficient reproduces $\chi(Y) = 27$;
6. orders $s > 6$ vanish dimensionally.

P01B.2-CURV - Stage A: Exact Curved Homogeneous Reduction

Status: calculation in progress; this stage derives and cross-locks the exact equations but does not yet certify a healthy branch.
Prerequisites used: P01A-G6.1.1, TOP00, G00, G01.

1. Four-scale geometry

For

$$ds_{16}^2 = -N^2 dt^2 + a^2 d\Sigma_{3,\kappa}^2 + \sum_{A=c,L,R} b_A^2 ds^2_{\mathbb{CP}^2_A,\mathrm{FS},1},$$

define

$$H = N^{-1}\dot{a}/a, \qquad h_A = N^{-1}\dot{b}_A/b_A, \qquad \Psi_I = -\left(N^{-1}\dot{H}_I + H_I^2\right).$$

The spatial second fundamental form is block diagonal with multiplicities

$$(d_0, d_c, d_L, d_R) = (3, 4, 4, 4).$$

The Gauss curvature double form is

$$\boxed{\Omega = {}^{(15)}R + K \wedge\!\!\wedge K,}$$

with external intrinsic sectional curvature κ/a^2, Fubini-Study curvatures $b_A^{-2}\mathcal{R}_{\mathrm{FS},A}$, and cross-block entries $\Omega^{ij}{}_{ij} = H_I H_J$.

2. Exact double-form normalization

Let $\Omega^{\wedge k}$ be the $(2k, 2k)$ double-form exterior power, and let tr_Δ denote the sum of equal upper/lower mask components. The canonical Lovelock density on the slice is

$$\mathcal{L}_k[\Omega] = 2^k \,\mathrm{tr}_\Delta(\Omega^{\wedge k}).$$

This normalization reproduces

$$\mathcal{L}_k(K\,\delta^{ij}_{kl}) = \frac{d!}{(d-2k)!}K^k$$

in dimension d, and reproduces the frozen G00 values $\mathcal{L}_1[\mathbb{CP}^2] = 24/b^2$, $\mathcal{L}_2[\mathbb{CP}^2] = 192/b^4$.

3. Lapse constraint

The exact lapse equation is

$$\boxed{\mathscr{H} = \frac{1}{2}\sum_{k=0}^{7}\alpha_k\mathcal{L}_k[\Omega] = 0, \qquad \mathcal{L}_0 = 1.}$$

This equation contains arbitrary $\alpha_0, \ldots, \alpha_7$, no normal acceleration, and no local α_8 term.

4. Four evolution equations

For a representative spatial index i_I in block I, the exact equation is

$$\boxed{\mathscr{S}_I = \sum_{k=0}^{7}\alpha_k\left[\mathscr{Q}^{(k)i_I}{}_{i_I} + \sum_{J=0,c,L,R}\mathcal{M}^{(k)}_{IJ}\Psi_J\right] = 0,}$$

where

$$\mathscr{Q}^{(k)i}{}_{j} = -\frac{1}{2^{k+1}}\delta^{ia_1b_1\cdots a_kb_k}_{jc_1d_1\cdots c_kd_k}\prod_{r=1}^{k}\Omega^{c_rd_r}{}_{a_rb_r},$$

$$\mathcal{A}^{ia}_{(k)jb} = \frac{k}{2^{k-1}}\delta^{iaa_2b_2\cdots a_kb_k}_{jbc_2d_2\cdots c_kd_k}\prod_{r=2}^{k}\Omega^{c_rd_r}{}_{a_rb_r},$$

and the four-scale acceleration block is

$$\boxed{\mathcal{M}^{(k)}_{IJ} = \sum_{a\in\text{block }J}\mathcal{A}^{i_Ia}_{(k)\,i_Ia}.}$$

These coefficient-extraction formulas are the explicit exact equations: the companion engine expands them with integer/rational arithmetic for any symbolic order or exact rational witness. They retain the full Fubini-Study Riemann tensor, not a constant-sectional-curvature replacement for $\mathbb{CP}^2$.

5. Complete homogeneous acceleration map

The physical regularity test is not $\det\mathcal{M}\neq 0$ alone. The inherited G6 map is

$$\boxed{\mathcal{A} = \sum_{k=1}^{7}\alpha_k\mathcal{A}_{(k)} : \mathrm{Sym}^2(T^*\Sigma^{15}) \longrightarrow \mathrm{Sym}^2(T^*\Sigma^{15}), \qquad \dim = 120.}$$

The exact engine constructs every component $\mathcal{A}^{ia}{}_{jb}$ and its 120×120 component-basis matrix. P01B.2 may pass only if this complete map, after the appropriate gauge/constraint interpretation, has no forbidden characteristic direction on the candidate branch.

6. Fixed-point coefficient system

At a static-radius de Sitter product,

$$H = H_\star > 0, \qquad h_A = 0, \qquad x = H_\star^2, \qquad y_A = b_{A\star}^{-2},$$

the four equations are linear in $\alpha_0,\ldots,\alpha_7$:

$$\boxed{\sum_{k=0}^{7}\alpha_k E_{I,k}(x,y_c,y_L,y_R) = 0.}$$

Their exact coefficients are ordinary-multinomial products of

$$\ell_{\mathrm{dS}_4} = (1, 12x, 24x^2), \qquad \ell_{\mathrm{dS}_3\perp} = (1, 6x),$$

$$\ell_{\mathbb{CP}^2_A} = (1, 24y_A, 192y_A^2), \qquad \ell_{\mathbb{CP}^2_A\perp} = (1, 12y_A).$$

For the external equation, the factors are $\mathrm{dS}_3\times(\mathbb{CP}^2)^3$; for the internal A equation, they are $\mathrm{dS}_4\times\mathbb{CP}^2_{A\perp}\times\prod_{B\neq A}\mathbb{CP}^2_B$. The common coefficient is $-1/2$.
The double-form dynamic equations and these compact fixed-point products agree order by order exactly in the companion regression.

7. P01B2A.1.2 frozen $\rho = 1$ controlled-slice audit

The numerical audit fixes

$$H_\star^2 = y_c = y_L = y_R = 1, \qquad y_A = b_{A\star}^{-2},$$

and therefore fixes the independent dimensionless ratio

$$\boxed{\rho = \frac{H_\star^2}{y} = 1.}$$

It is an equal-internal-radius, unit-curvature-ratio slice, not the complete equal-radius branch. In the four block-value basis the acceleration matrix M is self-adjoint with respect to

$$D = \mathrm{diag}(3,4,4,4), \qquad DM = M^T D.$$

The correct Euclidean symmetric representative and trace-sector spectrum are

$$\boxed{B = D^{1/2} M D^{-1/2} = B^T,} \qquad \mathrm{spec}(B_{\mathrm{Einstein}}) = \{-1,-1,-1,14\}.$$

P01B2A.1.2 uses eigvalsh(B), never eigvalsh(M). The first derivatives are constructed exactly before the final numerical spectrum by

$$\boxed{\partial_v\left(\Omega^{\wedge k}\right) = k\,\Omega^{\wedge(k-1)} \mathbin{\wedge\!\wedge} \partial_v \Omega.}$$

The calculation enforces a nonzero constraint-gradient norm, a maximum four-scale condition number, and a maximum constraint-tangent leakage. Its necessary homogeneous kinetic-sign diagnostic is the Einstein acceleration inertia $(1+, 119-)$; a physical ghost verdict is deferred until the constraint- and gauge-reduced tensor, radion, and vector residues are known.

The corrected deterministic sample contains

$$\text{4,000,000 coupling vectors,} \qquad \text{249,706 nontrace-inertia feasible points,}$$

$$\text{249,573 points passing all enforced necessary conditions,} \qquad \text{0 stable points.}$$

The invalid v1 trace operation omitted 27 legitimate trace-inertia passes; exact constraint-tangent differentiation shows that all 27 are unstable. The best sampled growth exponent remains positive,

$$\lambda_{\max} = 0.644082166066\ldots > 0.$$

This is a finite-sample no-pass statement on $\rho = 1$, not a no-go theorem.

8. Explicit topological order

For the order-eight Euler density, define on each time boundary the second fundamental-form one-form $\mathcal{K}^a = -K^a{}_b e^b$, with the sign chosen to match the frozen G6 positive-expansion convention. The Euler transgression is

$$\boxed{Q_{15} = 16 \int_0^1 ds\, \varepsilon_{a_1\cdots a_{15}}\, \mathcal{K}^{a_1} \wedge \prod_{r=1}^{7} \left({}^{(15)}R^{a_{2r}a_{2r+1}} - s^2 \mathcal{K}^{a_{2r}} \wedge \mathcal{K}^{a_{2r+1}} \right),}$$

and

$$\boxed{\mathcal{L}_8\, \mathrm{vol}_{16} = dQ_{15}.}$$

Thus

$$S_8 = \frac{\alpha_8}{2\kappa_{16}^2} \left[\int_{\Sigma_{t_2}} Q_{15} - \int_{\Sigma_{t_1}} Q_{15} \right]$$

up to spatial/asymptotic boundary pieces. In the flat Bianchi-I limit,

$$Q_{15} = 16\,14!\, V e_{15}(H), \qquad \mathcal{L}_8 = \frac{16\,14!}{V} \frac{d}{d\tau} \left(V e_{15} \right),$$

reproducing the independent P01B.1 audit identity. Therefore α_8 is retained at action level but cannot alter the four bulk equations or the G6 acceleration matrix. The companion exact engine now executes this identity and the isotropic oracle $16\,14!\,15 H^{16} = 16! H^{16}$ as mandatory regressions.

9. Stage-A verification and remaining gate

The exact engine checks:

1. all seven local orders against the flat P01B symmetric-polynomial equations;
2. all four fixed-point equations against independent product formulas;
3. construction of the complete 120×120 map;
4. the Einstein limit $(\mathcal{A}\Psi)^i{}_j = \delta^i_j \operatorname{tr}\Psi - \Psi^i{}_j$.

Stage A does **not** yet authorize P01C-CURV. The remaining P01B.2 work is a general-ratio/equal-radius regression followed by an unequal-radius full-order branch search, complete determinant margins, reduced physical residues, and stability tests. No branch will be called physically ghost-free from the four fixed-point equations or homogeneous acceleration inertia alone.

KKK-LL Canonical Project Memory - Roadmap v3 after P01B2A.1.2

Governing roadmap: KKK-LL Harmonised Master Roadmap - P01B.1 No-Go Revision v3
Effective date: 10 August 2026
Supersedes: P01B2A v1, P01B2A.1, P01B2A.1.1, and `PROJECT_MEMORY_ROADMAP_v3_TOP00_G01_P01B2A_1.md` for forward dependency and immutable-parent decisions.

Roadmap authority lock: the supplied v3 SVG, JSON dependency registry, and Markdown dependency matrix are canonical. The supplied PNG is visual-only because its byte hash does not match the accompanying checksum ledger.

Frozen scientific ancestry and release disposition

$$\boxed{\text{P01A}-\text{G6.1.1 FROZEN} \longrightarrow \text{P01B.1}-\text{FLAT COMPLETE/SCOPED NO}-\text{GO} \not\longrightarrow \text{P01C}-\text{FLAT.}}$$

P01B.1 is the all-local-order theorem for $\alpha_0, \ldots, \alpha_7$; its displayed fixed point and trajectory are EGB regressions. The topological $\alpha_8\mathcal{L}_8$ term has no bulk field tensor in $D = 16$.

The theorem is scientifically accepted for that scope, but original archive SHA-256 `036849...30787f2` is not release-freeze-grade because its verifier was manifest-optional and ignored generated bytecode residue. P01B.1.1 closed those defects but remained transcript-semantically fail-open. P01B.1.2 reseals the unchanged scientific calculation with mandatory manifest authentication, pre-import bytecode suppression, exact-tree checks, a pinned runtime, archive-level replay, and exact nested transcript-semantic authentication. Its active ancestor SHA-256 is `02c716bddc7ca0b08c220bd16104b7337a1e5df261360281405c4de61e952084`.

Completed curved-branch prerequisites

TOP00 - completed

The Cauchy topology for P01B.2-CURV is fixed before initial data:

$$\Sigma^{15} = \mathcal{X}^3_{\kappa,\Gamma} \times \mathbb{CP}^2_c \times \mathbb{CP}^2_L \times \mathbb{CP}^2_R.$$

Classical smooth globally hyperbolic evolution changes the metric, not this diffeomorphism type. P01B.2-CURV tests stabilization on a prescribed topology and homogeneous product ansatz; it does not claim topology creation or an unlabelled $3 + 12$ emergence.

G01 - completed for the P01B.2 prerequisite scope

The unequal-radius product tensors and integrated Lovelock panel $C_0, \ldots, C_6$ are exact and reproduce the equal-radius C00 values. The top internal coefficient satisfies

$$C_6 = (4\pi)^6 6!\, \chi((\mathbb{CP}^2)^3), \qquad \chi((\mathbb{CP}^2)^3) = 27.$$

P01B.2-CURV state after P01B2A.1.2

Exact Stage-A reduction - passed and unchanged scientifically

Completed:

1. exact lapse constraint for all $\alpha_0, \ldots, \alpha_7$;
2. four exact homogeneous block equations in G6 double-form coefficient form;
3. complete 120×120 homogeneous acceleration map;
4. fixed-point coefficient equations for $\mathrm{dS}_4 \times (\mathbb{CP}^2)^3$;
5. order-eight Euler transgression $\mathcal{L}_8\mathrm{vol} = dQ_{15}$, now connected to an executable total-derivative and isotropic regression;
6. exact flat-P01B, compact-product, Einstein, and all-order regressions.

P01B2A v1 - held and superseded

The v1 numerical trace audit used `eigvalsh(M)` on a nonsymmetric block-value matrix. The block dimensions are $(3, 4, 4, 4)$, so

$$D = \mathrm{diag}(3, 4, 4, 4), \qquad DM = M^T D, \qquad B = D^{1/2} M D^{-1/2} = B^T.$$

P01B2A.1 corrected the scientific trace operation by using eigvalsh(B) and enforcing the Einstein oracle

$$\mathrm{spec}(B_E) = \{-1, -1, -1, 14\}.$$

The v1 release is historical evidence only and is not an immutable parent. P01B2A.1 is scientifically accepted but its archive is also historical HOLD provenance because of active-text, transcript, and ancestor-envelope defects. P01B2A.1.1 closed the earlier release defects without altering the scientific values, but its stored transcript did not semantically enforce nested stage statuses, counts, byte identities, or archive cross-locks. P01B2A.1.2 closes that final release defect and is the only authorized immutable parent for the next calculation.

Equal-internal-radius, $\rho = 1$ controlled slice - corrected no pass

The audited point is

$$H_\star^2 = y_c = y_L = y_R = 1, \qquad y_A = b_{A\star}^{-2}, \qquad \rho = \frac{H_\star^2}{y} = 1.$$

This is one unit-curvature-ratio slice of the equal-radius branch. It is not the complete equal-radius family. Two exact on-shell nonzero-all-order witnesses, combined with exact first derivatives of the reduced equations, show:

1. one constraint-tangent stable witness is full rank and has positive external Planck coefficient but fails the necessary Einstein-sign acceleration-inertia condition;
2. one full-rank witness passes the homogeneous $(1+, 119-)$ Einstein-inertia condition and has positive Planck coefficient but has growing constraint-tangent radion modes.

These are necessary homogeneous kinetic-sign diagnostics. A physical ghost verdict is deferred to the constraint- and gauge-reduced tensor, radion, and vector residues.

The corrected deterministic scan of 4,000,000 coupling vectors found:

$$\text{249,706 nontrace-inertia feasible points,} \qquad \text{249,573 points passing all enforced necessary conditions,}$$

$$\text{0 stable joint points,} \qquad \lambda_{\max}^{\mathrm{best}} = 0.644082166066\ldots > 0.$$

The invalid v1 trace operation omitted 27 genuine trace-inertia passes. Exact constraint-tangent differentiation includes all 27 and finds them all unstable, with

$$1.6151308423\ldots \le \lambda_{\max} \le 1.8686083466\ldots.$$

This is a finite-sample no-pass result on $\rho = 1$. It is not an all-coupling no-go theorem, does not cover arbitrary ρ, and does not cover unequal radii.

Authorized next work

The next calculation remains inside P01B.2-CURV:

$$\boxed{\begin{array}{r}\text{arbitrary-}\rho\text{ equal-radius regression} \longrightarrow \text{unequal-radius continuation} \\ \longrightarrow \text{reduced residues, determinant margins, and stability}\end{array}}$$

It must vary $(H_\star^2, y_c, y_L, y_R)$, or independent ratios $\rho_A = H_\star^2/y_A$, without fixing all four scales to unity. It must retain all local Lovelock orders, the action-level α_8 ledger, weighted trace normalization, exact first derivatives, the complete 120-mode map, and physical residue reduction.

P01C-CURV remains blocked until one branch passes every P01B.2 health gate. If no such branch exists, the roadmap handoff is P01B.3-GEN or P01B.M-FULL, not P01C-CURV.

Canonical branch map

PRIMARY :	$\mathrm{TOP00} \to \mathrm{G01} \to \mathrm{P01B.2-CURV} \to \mathrm{P01C-CURV} \to \mathrm{P05} \to \mathrm{P06} \to \mathrm{P07},$
STRONG EMERGENCE :	$\mathrm{TOP00} \to \mathrm{P01B.3-GEN} \to \mathrm{P01C-GEN} \to \mathrm{P02} \to \mathrm{P03} \to \mathrm{P04},$
FULL ACTION :	$\mathrm{TOP00} \to \mathrm{P01B.M-FULL} \to \mathrm{P01C-FULL},$
OPTIONAL BOUNDARY :	$\mathrm{P01B.D-DAE}.$

No downstream matter, chirality, global lift, four-dimensional EFT, cosmology, bag, or Zeptometry claim is authorized before a healthy P07 branch. Z00 may continue only as a symbolic interface.